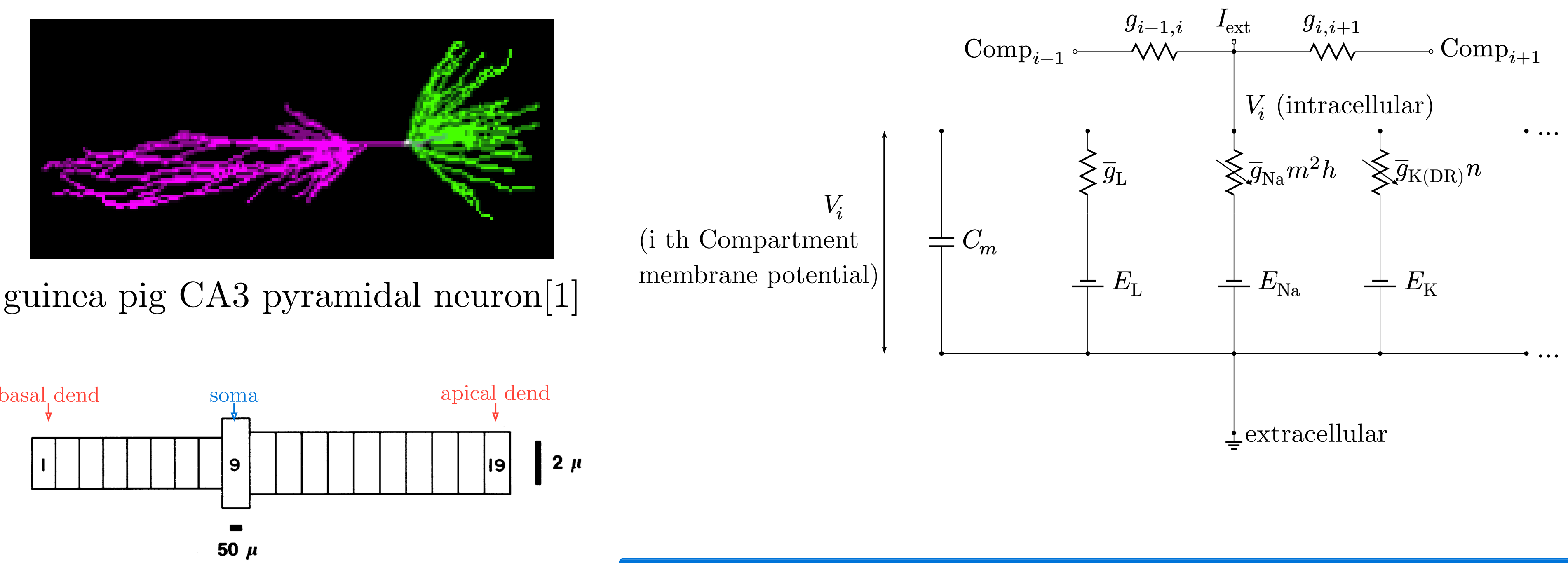


Introduction



Conductances have **gate variables**, ODEs with rate functions α, β :

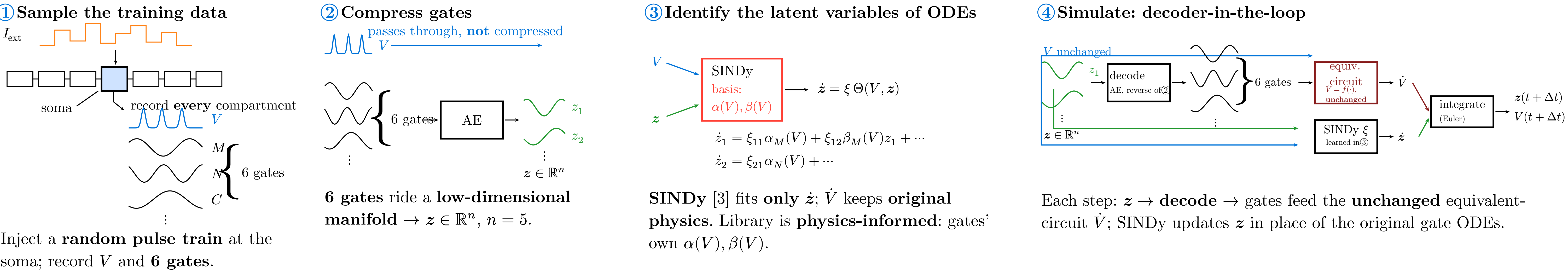
$$I_{Na} = \bar{g}_{Na} m^2 h (V - E_{Na})$$
$$\frac{dm}{dt} = \alpha_m(V)(1 - m) - \beta_m(V)m$$
$$\alpha_m(V) = 0.32(13.1 - u)/(\exp((13.1 - u)/4) - 1)$$
$$\beta_m(V) = 0.28(u - 40.1)/(\exp((u - 40.1)/5) - 1)$$

→ **11 states per compartment**
→ **209** for 19 comps;
At large scale network simulation, the gates become a **memory bottleneck**.

Goal

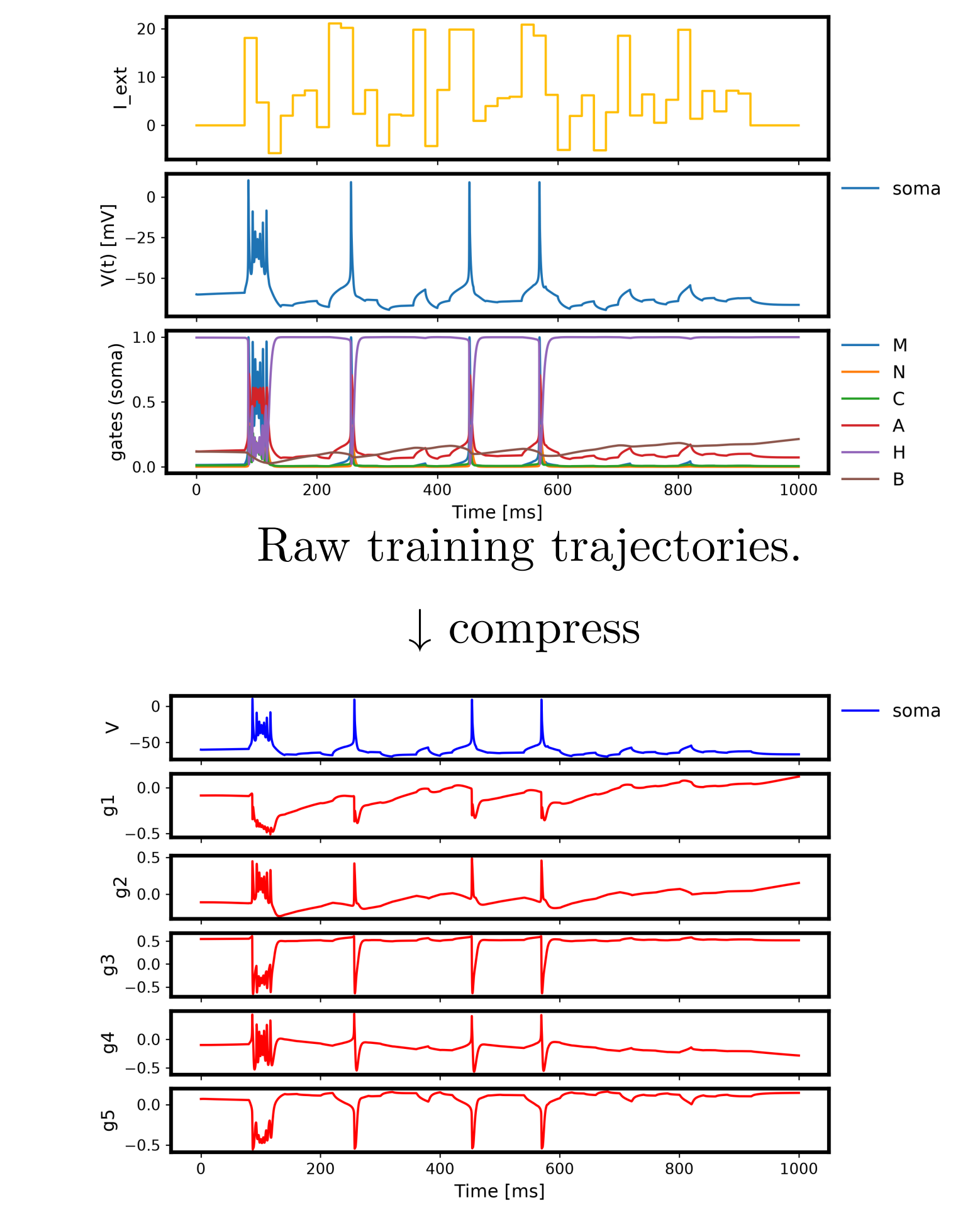
Development of a multi-compartment neuron’s surrogate model capable of reproducing the membrane potential response with few gate variables.

Methods

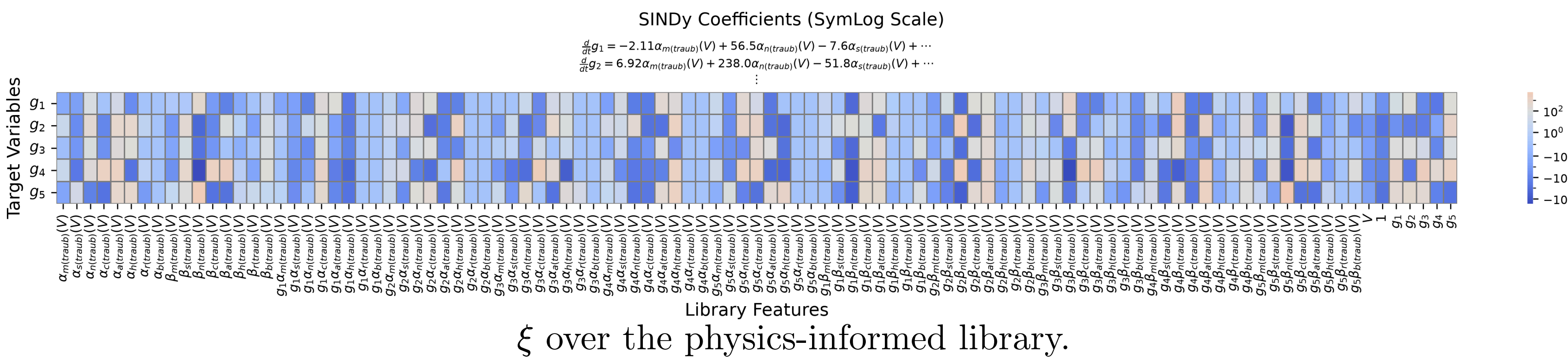


Results and Discussion

Training Data to capture gate dynamics

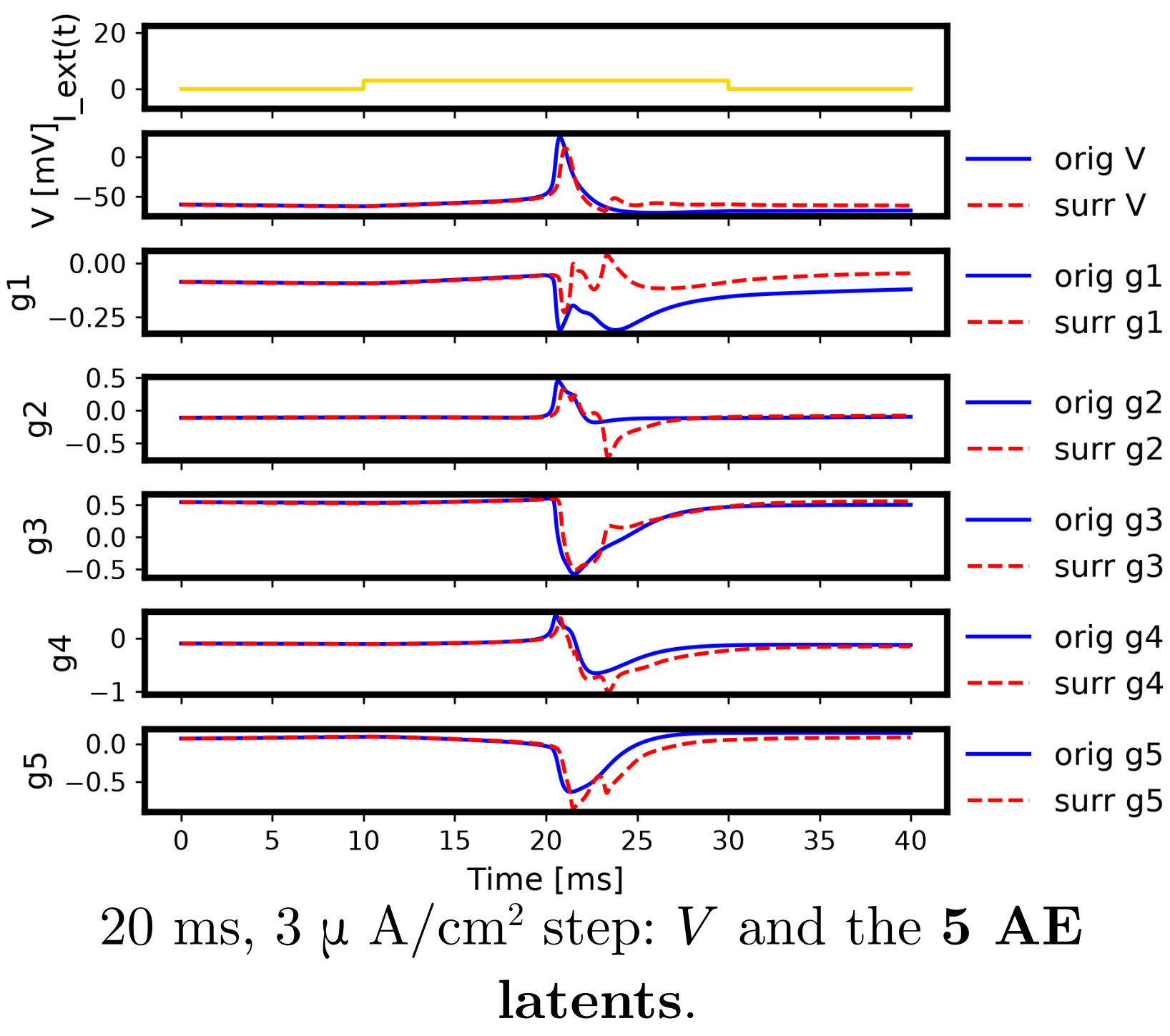


Latent trajectories used to fit the SINDy library.
Identified latent equations



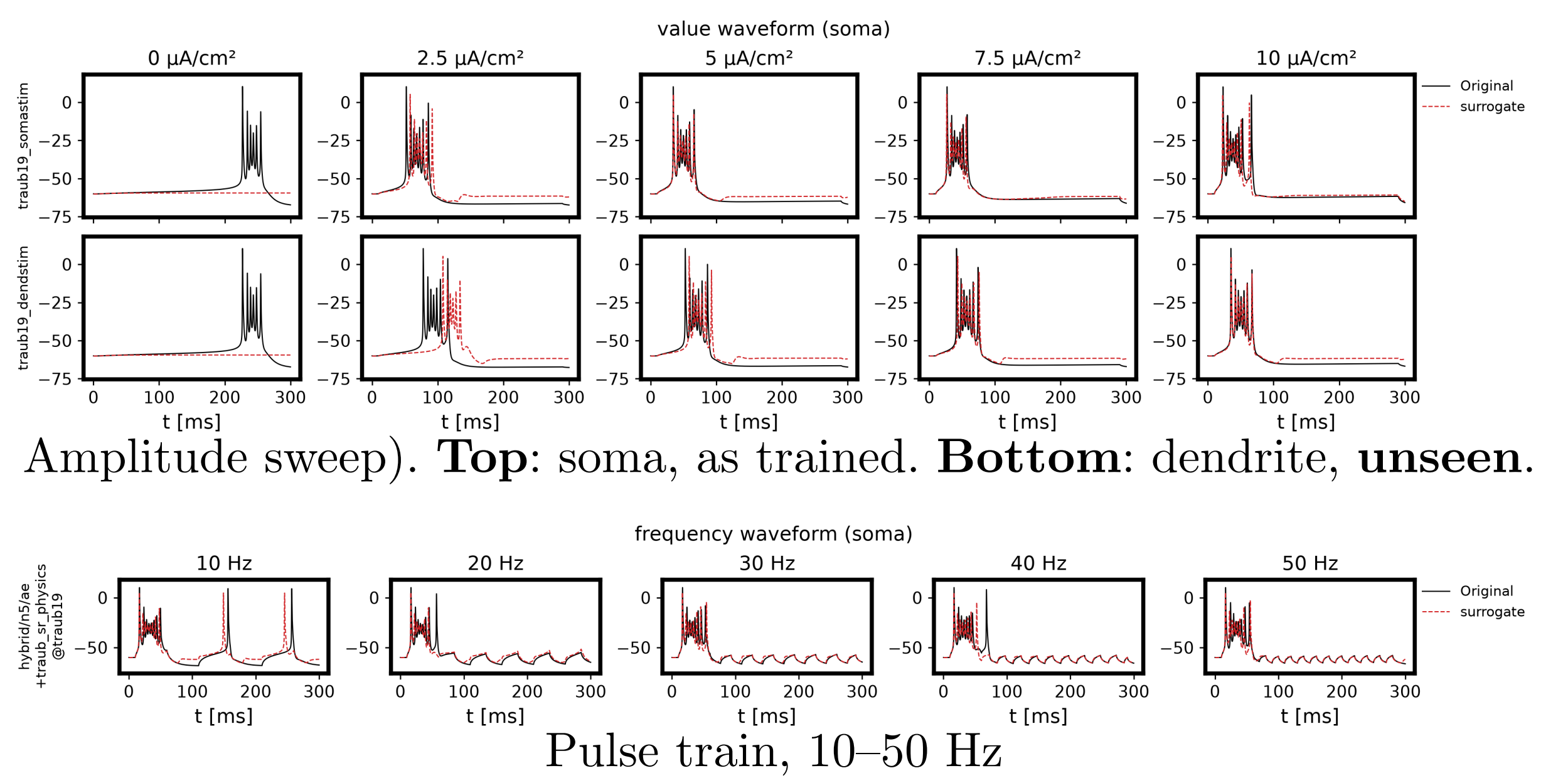
- **79.6% non-zero** — accurate but **not sparse**.

Action Potential reproduction

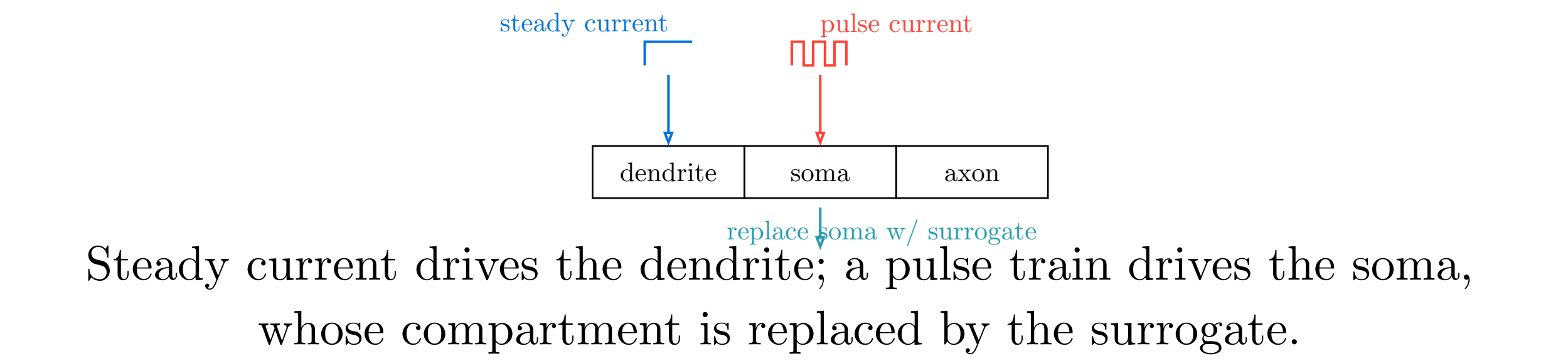


- latency err **0.3 ms**, AHP timing gap **3.1 ms**.
- peak **13 mV** low (amplitude gap **12.9 mV**), rise / fall rate gap **21.0 / 10.3 mV/ms** — yet AHP depth gap is only **0.28 mV**.

② New stimulus



- Bursts reproduced at **both sites** for $I \geq 5$; fires **too early** near threshold.
 - Matches for $f \geq 20$ Hz; **drops later spikes** at 10 Hz.
- Stimulus and replacement sites**



Conclusion

- Latent dynamics preserve **timing**.
- Error concentrates in the **fast, steep transition** — suggests the **AE encoding (n=5)**, not the SINDy library, fails to preserve it.
- Replaced **all 19 compartments**, no divergence; transfers to **unseen site • frequency**.
- **Core finding**: SINDy succeeds **not because we compressed (n: 6→5)** —⑤shows the **same result uncompressed (n = 6)**, so it is the **coordinate change itself** that makes gate dynamics learnable.
- Cost reduction follows **only if** compression matters — so it remains **unproven**: ξ is still dense (**79.6%** non-zero).
→ **Future**: sparsify ξ , push n down, test on other channel models.